RESEARCH REVIEW

AI Models to Detect Dryness

Computer Vision, Deep Learning, Smart Sensors, and Solar Drying

Project focus: AI-based detection of dryness in leaves and agricultural products using images and sensor data.

Prepared as a literature-review starting point

Note: Verify each paper's metadata and reported results against the publisher's full text before academic submission. This review does not claim that the selected papers are the five newest publications as of October 2026.

1. Introduction

Dryness detection is important in agriculture, food processing, and solar drying. Insufficient drying can contribute to spoilage, while excessive drying may reduce product quality. Traditional approaches include manual inspection, weighing, and moisture meters. AI can help estimate the drying stage by analysing images, temperature, relative humidity, weight change, and drying time.

A proposed smart system may combine computer vision for colour and texture analysis, machine learning for prediction, smart sensors for environmental measurements, wireless connectivity for monitoring, and edge computing for local inference.

Important: Visual appearance alone cannot reliably establish actual moisture content. AI predictions should be validated against a suitable reference moisture measurement.

2. Five Relevant Research Papers

Paper 1 — AI-Driven Optimization of Solar Drying for Cooked Rice

Publication details: Authors reported in the supplied publisher record: M. Rajeswaran, K. Senthil Kumar, B. Siddharthan, R. Arunkumar, R. Bhoopathi, S. P. Krishnendhu, and Arunkumar Munimathan. Publisher page lists publication date as 15 July 2026; verify full citation and article details before submission.

Methods: Methods described in the publisher record include IoT measurements (temperature, humidity, solar irradiance, airflow), multiple predictive models, U-Net–ConvLSTM, and explainable-AI techniques such as SHAP.

Observation and research gap: Observation: The work illustrates how time-varying environmental data can support drying prediction and optimization. Research opportunity: test generalization from cooked rice to leaves and validate against measured moisture content.

Paper link: https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70557?af=R

Paper 2 — Enhancing Post-Harvest Sustainability in Temperate Crops Through a Smart IoT-Integrated Indirect Solar Dryer

Publication details: Authors include Brij Bhushan Sharma, Pankaj Vaidya, Nitin Kumar, Ankita Tiwari, Shonak Bansal, Mohammad Rashed Iqbal Faruque, and colleagues. Scientific Reports publisher page, 2025.

Methods: Methods described include an indirect solar dryer, IoT-based temperature/airflow monitoring, remote monitoring, and crop-specific drying conditions.

Observation and research gap: Observation: IoT can improve monitoring and control. Research opportunity: add a validated image/sensor AI model that estimates product moisture, rather than relying on chamber conditions alone.

Paper link: https://www.nature.com/articles/s41598-025-13499-x

Paper 3 — Performance Evaluation of a Solar Hybrid Collector With Dryer Using Experimental and Machine Learning Approaches

Publication details: The Scientific Reports publisher record is associated with DOI/article identifier 10.1038/s41598-025-29714-8; verify the final citation, author list, and publication year from the publisher page.

Methods: The described study compares open-sun, natural-convection, and forced-convection drying and evaluates Random Forest and Support Vector Regression for predicting drying performance.

Observation and research gap: Observation: Experimental drying measurements can be paired with regression models. Research opportunity: add camera-based leaf features and evaluate performance on separate drying batches.

Paper link: https://www.nature.com/articles/s41598-025-29714-8

Paper 4 — Moisture Content Online Detection System Based on Multi-Sensor Fusion and Convolutional Neural Network

Publication details: A related open-access record was provided through PubMed Central. Confirm exact title, author list, journal, and year from the full-text record before citing.

Methods: Methods described include multi-sensor data collection, sensor fusion, and CNN-based moisture prediction.

Observation and research gap: Observation: Multiple sensor inputs can represent the drying state better than a single environmental reading. Research opportunity: combine sensor readings with RGB images and test on different leaf varieties.

Paper link: https://pmc.ncbi.nlm.nih.gov/articles/PMC10944943/

Paper 5 — Prediction and Visualization of Moisture Content in Tencha Drying Processes by Computer Vision and Deep Learning

Publication details: PubMed record PMID 38349009. Confirm the full bibliographic metadata and exact methods from the article before submission.

Methods: The described approach uses image-derived colour features and compares a 1D-CNN, Partial Least Squares, and a backpropagation neural network, with preprocessing methods including Z-score normalization.

Observation and research gap: Observation: Image features can help estimate moisture during tea processing. Research opportunity: test robustness under changing lighting and transfer to other leaf types.

Paper link: https://pubmed.ncbi.nlm.nih.gov/38349009/

3. Comparative Summary

No.	Research focus	Methods / technology	Use for this project
1	Solar drying of cooked rice	IoT, U-Net–ConvLSTM, explainable AI	Temporal drying prediction
2	Smart indirect solar dryer	IoT monitoring and control	Hardware/connectivity design
3	Solar drying of mint leaves	Random Forest, SVR	Experimental data and model comparison
4	Online moisture detection	Sensor fusion, CNN	Multisensor prediction
5	Tencha tea drying	Computer vision, 1D-CNN, PLS, ANN	Image-based moisture estimation

4. Overall Observations

1. AI can estimate drying state. Machine learning can model relationships between measured features and reference moisture content, provided representative data are available.

2. Colour and texture are useful but not sufficient by themselves. Lighting, plant variety, camera settings, and leaf thickness can affect image appearance.

3. Sensor fusion is promising. Images, temperature, humidity, and weight change provide complementary information about drying.

4. Time-series models may capture drying progression. CNN-LSTM or ConvLSTM can be considered when enough sequential data are available; simpler models should be used as baselines.

5. Connectivity supports practical deployment. IoT enables monitoring and alerts, but the system needs local fail-safe controls and validated stopping criteria.

5. Research Gaps

Generalization across leaf types: Train and evaluate on multiple varieties and keep separate batches for testing.

Lighting changes: Use controlled lighting where possible and test normalization under sunlight and shadows.

Single-input limitations: Fuse images with temperature, relative humidity, and weight measurements.

Reliable stopping point: Use reference moisture measurements to define the target threshold.

Real-time deployment: Run inference on a laptop or edge device and connect it to a dashboard.

Weather variation: Collect data across different solar and ambient conditions.

Data leakage and weak validation: Split by drying batch/day rather than randomly splitting near-duplicate images.

Explainability: Display predicted moisture, confidence, and relevant input features.

6. Proposed Innovation

Proposed title: AI-Based Leaf Dryness Detection Using Computer Vision and Smart Sensor Integration in a Solar Dryer.

The proposed system will capture leaf images, collect temperature and humidity readings, and measure sample weight during drying. A model will estimate moisture content or classify the sample as wet, partially dry, or sufficiently dry. Results will be displayed on a dashboard, with an alert when a validated target is reached.

Potential contributions include (1) image-based dryness estimation, (2) multimodal prediction, (3) real-time monitoring, (4) controlled alerts or shutdown, and (5) a low-cost prototype. These are proposed project features, not claims that the five papers already implemented them.

7. Proposed Hardware and Connectivity

Component	Purpose
RGB camera	Capture leaf images at regular intervals.
Temperature sensor	Measure drying-chamber temperature.
Relative-humidity sensor	Measure chamber/ambient humidity.
Load cell + amplifier	Track mass change during drying.
ESP32	Collect readings and communicate with the processing device.
Raspberry Pi or laptop	Run the AI model and dashboard.
Wi-Fi / MQTT / USB serial	Transfer readings and predictions.
Display / dashboard	Show sensor readings, predicted moisture, and status.

Suggested prototype connectivity: Start with sensors connected to an ESP32 and a laptop/Raspberry Pi. Use USB/serial for early experiments, then add Wi-Fi/MQTT for remote monitoring. If the model controls a physical dryer, use appropriate electrical isolation and independent safety cut-offs.

8. Recommended AI Methods

Method	Purpose	Recommendation
Linear Regression	Numerical baseline for moisture prediction.	Start here for a simple baseline.
Random Forest	Predict moisture from sensor and extracted image features.	Good starting model for tabular data.
SVR	Continuous moisture prediction.	Compare with Random Forest.
CNN	Learn visual patterns from images.	Use if enough labelled images are available.
CNN-LSTM / ConvLSTM	Model changes in images/features over time.	Advanced option after a baseline works.

Evaluation: use MAE, RMSE, and R^2 for continuous moisture prediction. For wet/partially dry/dry classification, use accuracy, precision, recall, F1-score, and a confusion matrix. Test on separate drying batches or days to avoid

overestimating performance.

9. Reference Moisture Measurement

A reference moisture measurement is necessary to train and evaluate the AI model. One common wet-basis formula is:

Moisture content (wet basis, %) = $((m_{sample} - m_{dry}) / m_{sample}) \times 100$

Here, m_sample is the sample mass at the relevant measurement time, and m_dry is the dry mass determined using an appropriate validated reference procedure. Choose a method suitable for the material and document it consistently.

10. How to Read the Papers

1. Read the title and abstract to identify the problem, product, method, and key result.
2. Read the introduction to understand the motivation and prior limitations.
3. Study the methodology: dataset, sensors, image processing, model, training/test split, and output.
4. Inspect results and note the metrics; check whether the test data are independent.
5. Read the conclusion and limitations, separating author-stated limitations from your own observations.
6. Record each paper in a literature table: year, inputs, model, output, results, limitations, and proposed improvement.
7. Compare papers and state one clear gap your project will test.

11. Research Observations — Ready-to-Use Summary

The reviewed research directions show that AI, computer vision, and smart sensors can support agricultural drying. Machine learning can estimate drying performance from experimental data, while computer vision can use colour and texture changes to estimate moisture-related properties. IoT can support continuous data collection and remote monitoring, but temperature and humidity alone do not confirm the moisture content inside a leaf.

A promising extension is to combine leaf images with temperature, relative humidity, and weight measurements. The proposed system should be trained against reference moisture measurements and evaluated on independent drying batches and, where possible, multiple leaf varieties. This can help determine whether multimodal input improves reliability under changing solar and environmental conditions.

12. References and Verification Notes

1. AI-Driven Optimization of Solar Drying for Cooked Rice — publisher page: https://aiche.onlinelibrary.wiley.com/doi/10.1002/ep.70557?af=R

2. Enhancing Post-Harvest Sustainability in Temperate Crops Through a Smart IoT-Integrated Indirect Solar Dryer — Scientific Reports: https://www.nature.com/articles/s41598-025-13499-x

3. Performance Evaluation of a Solar Hybrid Collector With Dryer Using Experimental and Machine Learning Approaches — Scientific Reports: https://www.nature.com/articles/s41598-025-29714-8

4. Moisture Content Online Detection System Based on Multi-Sensor Fusion and Convolutional Neural Network — PubMed Central record: https://pmc.ncbi.nlm.nih.gov/articles/PMC10944943/

5. Prediction and Visualization of Moisture Content in Tencha Drying Processes by Computer Vision and Deep Learning — PubMed record PMID 38349009: https://pubmed.ncbi.nlm.nih.gov/38349009/

Academic integrity note: Before submitting, open each source and verify the exact title, author list, publication year, journal, DOI, methods, and numerical results. The references above are starting points; this document is a research-review draft and should not be represented as proof that all five are the latest papers available.